PROJECT CASE STUDY

Cobot Visuomotor Dataset Pipeline

Video-Sensor Synchronization for Imitation Learning

Category	Status	Year
Data Engineering	Completed pipeline	2026

An automated pipeline that trims inactive robot data, synchronizes video frames with timestamps, and produces image-to-joint targets for cobot imitation learning.

76	10 FPS	6
aligned samples	extraction rate	joint targets

Context and problem

Raw cobot logs and video are not immediately trainable: movement must be isolated, frames must align with sensor time, and every image needs a consistent joint-angle target.

Strategy and implementation

- Detected the last meaningful six-joint movement and trimmed inactive data.
- Mapped every retained timestamp to its exact video frame.
- Extracted synchronized images at 10 FPS.
- Compiled a JSON dataset containing image path, timestamp, and six target joint angles.

My role

Designed and implemented the complete data-cleaning, synchronization, extraction, and dataset-building workflow.

Technology stack

Python | OpenCV | Pandas | JSON | Robot Joint Logs | Video Processing

Results and evidence

- Generated 76 synchronized image-joint samples.
- Covered timestamps from 0.0 to 7.5 seconds.
- Produced a cross-platform JSON manifest ready for visuomotor learning experiments.

Key takeaways

- Reliable synchronization is the core of multimodal robotics data.
- A dataset pipeline should validate one-to-one correspondence before model training.

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